

Note 8.4 Integration of Rational Functions by Partial Fractions

The Method of Partial Fractions

- **Definition:**

Express a rational function as a sum of simpler fractions and the method for rewriting rational functions as a sum of simpler fractions is the method of partial fractions.

- **Example:**

$\frac{5x - 3}{x^2 - 2x - 3} = \frac{A}{x + 1} + \frac{B}{x - 3}$. We call A and B is undetermined coefficients.

General Description of the Method

- **Conditions:**

For the rational function $f(x)/g(x)$, if we want to use the method,

- The degree of $f(x)$ must be less than the degree of $g(x)$. For the another case, we can turn it into this case by divide the rational function into the sum of [the proper rational function](#) and a polynomial function.
- We must know the factors of $g(x)$.

- **Method of Partial Fractions ($\frac{f(x)}{g(x)}$ Proper)**

- Let $x - r$ be a linear factor of $g(x)$, Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$, then to this factor, we can assign the sum as

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \dots + \frac{A_m}{(x - r)^m}$$

And we need to do this for each distinct linear factor of $g(x)$.

- For the irreducible quadratic factor of $g(x)$, the method is the same but the numerator is a linear function of x .
- Then we need to express the original fraction as the sum of all the partial fractions and arrange the terms in decreasing powers of x .
- Solve the undetermined coefficients.

The Heaviside "Cover-up" Method for Linear Factors

- **Condition:**

$g(x)$ can be expressed as different the linear factors.

- **Heaviside Method:**

- Write the quotient with $g(x)$ factored:

$$\frac{f(x)}{g(x)} = \frac{f(x)}{(x - r_1)(x - r_2) \cdots (x - r_n)}$$

- Cover the factors $(x - r_i)$ of $g(x)$ one at a time, each time replacing all the uncovered x 's by the number r_i . Then we can get A_i for each root r_i .

Exercises

- P479 37, 47, 49
- [P509 Question 2nd](#)

For this question, we need to divide it into the sum of $m + 1$ fractions, that is,

$$\frac{dx}{x(x+1)(x+2) \cdots (x+m)} = \frac{A_0}{x} + \frac{A_1}{x+1} + \cdots + \frac{A_m}{x+m}$$

Here, we can use the Heaviside Method to obtain the formula of A_k , which is

$$A_k = \frac{(-1)^k}{k(m-k)}$$

Thus

$$\int \frac{dx}{x(x+1)(x+2) \cdots (x+m)} = \sum_{k=0}^m \frac{(-1)^k}{k(m-k)} \ln(x+k) + C$$

- [P510 Question 6th](#)

First, we need to transform it into the form of the difference of two squares,

$$\int \frac{dx}{x^4 + 4} = \int \frac{1}{(x^2 + 2)^2 - (2x)^2} dx = \int \frac{1}{(x^2 - 2x + 2)(x^2 + 2x + 2)} dx$$

Then we can split the product of two fraction,

$$\frac{1}{(x^2 - 2x + 2)(x^2 + 2x + 2)} = \frac{Ax + B}{x^2 - 2x + 2} + \frac{Cx + D}{x^2 + 2x + 2}$$

Then we have

$$\frac{-\frac{1}{8}x + \frac{1}{4}}{x^2 - 2x + 2} + \frac{\frac{1}{8}x + \frac{1}{4}}{x^2 + 2x + 2} = \frac{1}{8} \left[\frac{-(x-1)+1}{(x-1)^2+1} - \frac{(x+1)+1}{(x+1)^2+1} \right]$$

Thus

$$\int \frac{1}{x^4 + 4} \, dx = \frac{1}{16} \ln \left| \frac{x^2 + 2x + 2}{x^2 - 2x + 2} \right| + \frac{1}{8} [\arctan(x + 1) + \arctan(x - 1)] + C$$